

SQL | MYSQL | FINANCIAL_LOAN TABLE

BANK LOAN REPORT

Query Document – KPIs, Loan Quality & Overview Dashboards

A1	KPIs	Applications, Funded Amount, Amount Received, Interest Rate, DTI (Total / MTD / PMTD)
A2	Good Loan Issued	Fully Paid + Current loans
A3	Bad Loan Issued	Charged Off loans
A4	Loan Status	Status-wise summary and MTD summary
B	Overview	Month, State, Term, Employee Length, Purpose, Home Ownership
+	Filter Example	Grade A filter applied to the Purpose query

PART A

BANK LOAN REPORT | SUMMARY

A1. KPIs

1.1 Total Loan Applications

SQL QUERY

```
SELECT COUNT(id) AS Total_Loan_Applications
FROM financial_loan;
```

OUTPUT

Total_Loan_Applications
38576

1.2 MTD Loan Applications

SQL QUERY

```
SELECT COUNT(id) AS MTD_Total_Applications
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

MTD_Total_Applications
4314

1.3 PMTD Loan Applications

SQL QUERY

```
SELECT COUNT(id) AS PMTM_Total_Applications
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 11
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

	PMTM_Total_Applications
▶	4035

2.1 Total Funded Amount

SQL QUERY

```
SELECT SUM(loan_amount) AS Total_Funded_Amount
FROM financial_loan;
```

OUTPUT

	Total_Funded_Amount
▶	435757075

2.2 MTD Total Funded Amount

SQL QUERY

```
SELECT SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

	MTD_Total_Funded_Amount
▶	53981425

2.3 PMTD Total Funded Amount

SQL QUERY

```
SELECT SUM(loan_amount) AS PMTD_Total_Funded_Amount
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 11
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

	PMTD_Total_Funded_Amount
▶	47754825

3.1 Total Amount Received

SQL QUERY

```
SELECT SUM(total_payment) AS Total_Amount_Collected
FROM financial_loan;
```

OUTPUT

Total_Amount_Collected
473070933

3.2 MTD Total Amount Received

SQL QUERY

```
SELECT SUM(total_payment) AS MTD_Total_Amount_Collected
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

MTD_Total_Amount_Collected
58074380

3.3 PMTD Total Amount Received

SQL QUERY

```
SELECT SUM(total_payment) AS PMTD_Total_Amount_Collected
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 11
AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

PMTD_Total_Amount_Collected
50132030

4.1 Average Interest Rate

SQL QUERY

```
SELECT AVG(int_rate)*100 AS Avg_Int_Rate
FROM financial_loan;
```

OUTPUT

Avg_Int_Rate
12.048831397760178

4.2 MTD Average Interest

SQL QUERY

```
SELECT AVG(int_rate)*100 AS MTD_Avg_Int_Rate
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

MTD_Avg_Int_Rate
12.356040797403738

4.3 PMTD Average Interest

SQL QUERY

```
SELECT AVG(int_rate)*100 AS PMTD_Avg_Int_Rate
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 11
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

PMTD_Avg_Int_Rate
11.941717472118796

5.1 Average DTI

SQL QUERY

```
SELECT AVG(dti)*100 AS Avg_DTI
FROM financial_loan;
```

OUTPUT

Avg_DTI
13.32743311903776

5.2 MTD Average DTI

SQL QUERY

```
SELECT AVG(dti)*100 AS MTD_Avg_DTI
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

MTD_Avg_DTI
13.66553778395922

5.3 PMTD Average DTI

SQL QUERY

```
SELECT AVG(dti)*100 AS PMTD_Avg_DTI
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 11
      AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021;
```

OUTPUT

PMTD_Avg_DTI
13.302733581164853

A2. GOOD LOAN ISSUED

1 Good Loan Percentage

SQL QUERY

```
SELECT
  (COUNT(CASE WHEN loan_status = 'Fully Paid'
                OR loan_status = 'Current' THEN id END) * 100.0)
  / COUNT(id) AS Good_Loan_Percentage
FROM financial_loan;
```

OUTPUT

	Good_Loan_Percentage
▶	86.17534

2 Good Loan Applications

SQL QUERY

```
SELECT COUNT(id) AS Good_Loan_Applications
FROM financial_loan
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';
```

OUTPUT

	Good_Loan_Applications
▶	33243

3 Good Loan Funded Amount

SQL QUERY

```
SELECT SUM(loan_amount) AS Good_Loan_Funded_amount
FROM financial_loan
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';
```

OUTPUT

	Good_Loan_Funded_amount
▶	370224850

4 Good Loan Amount Received

SQL QUERY

```
SELECT SUM(total_payment) AS Good_Loan_amount_received
FROM financial_loan
WHERE loan_status = 'Fully Paid' OR loan_status = 'Current';
```

OUTPUT

	Good_Loan_amount_received
▶	435786170

A3. BAD LOAN ISSUED

1 Bad Loan Percentage

SQL QUERY

```
SELECT
  (COUNT(CASE WHEN loan_status = 'Charged Off' THEN id END) * 100.0)
  / COUNT(id) AS Bad_Loan_Percentage
FROM financial_loan;
```

OUTPUT

	Bad_Loan_Percentage
▶	13.82466

2 Bad Loan Applications

SQL QUERY

```
SELECT COUNT(id) AS Bad_Loan_Applications
FROM financial_loan
WHERE loan_status = 'Charged Off';
```

OUTPUT

	Bad_Loan_Applications
▶	5333

3 Bad Loan Funded Amount

SQL QUERY

```
SELECT SUM(loan_amount) AS Bad_Loan_Funded_amount
FROM financial_loan
WHERE loan_status = 'Charged Off';
```

OUTPUT

	Bad_Loan_Funded_amount
▶	65532225

4 Bad Loan Amount Received

SQL QUERY

```
SELECT SUM(total_payment) AS Bad_Loan_amount_received
FROM financial_loan
WHERE loan_status = 'Charged Off';
```

OUTPUT

	Bad_Loan_amount_received
▶	37284763

A4. LOAN STATUS

1 Loan Status Summary

SQL QUERY

```
SELECT
  loan_status,
  COUNT(id) AS LoanCount,
  SUM(total_payment) AS Total_Amount_Received,
  SUM(loan_amount) AS Total_Funded_Amount,
  AVG(int_rate * 100) AS Interest_Rate,
  AVG(dti * 100) AS DTI
FROM financial_loan
GROUP BY loan_status;
```

OUTPUT

	loan_status	LoanCount	Total_Amount_Received	Total_Funded_Amount	Interest_Rate	DTI
▶	Charged Off	5333	37284763	65532225	13.878574910931917	14.004732795799695
	Fully Paid	32145	411586256	351358350	11.641070773058658	13.167350754394164
	Current	1098	24199914	18866500	15.0993260473588	14.724344262295068

2 Loan Status | MTD (December 2021)

SQL QUERY

```
SELECT
  loan_status,
  SUM(total_payment) AS MTD_Total_Amount_Received,
  SUM(loan_amount) AS MTD_Total_Funded_Amount
FROM financial_loan
WHERE MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 12
AND YEAR(STR_TO_DATE(issue_date, '%d-%m-%Y')) = 2021
GROUP BY loan_status;
```

OUTPUT

	loan_status	MTD_Total_Amount_Received	MTD_Total_Funded_Amount
▶	Fully Paid	47815851	41302025
	Charged Off	5324211	8732775
	Current	4934318	3946625

PART B

BANK LOAN REPORT | OVERVIEW

B1. MONTH

1 Month-wise Summary

SQL QUERY

```

SELECT
  MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')) AS Month_Number,
  MONTHNAME(STR_TO_DATE(issue_date, '%d-%m-%Y')) AS Month_Name,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY
  MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y')),
  MONTHNAME(STR_TO_DATE(issue_date, '%d-%m-%Y'))
ORDER BY
  MONTH(STR_TO_DATE(issue_date, '%d-%m-%Y'));

```

OUTPUT

	Month_Number	Month_Name	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	1	January	2332	25031650	27578836
	2	February	2279	24647825	27717745
	3	March	2627	28875700	32264400
	4	April	2755	29800800	32495533
	5	May	2911	31738350	33750523
	6	June	3184	34161475	36164533
	7	July	3366	35813900	38827220
	8	August	3441	38149600	42682218
	9	September	3536	40907725	43983948
	10	October	3796	44893800	49399567
	11	November	4035	47754825	50132030
	12	December	4314	53981425	58074380

B2. STATE

1 State-wise Summary

SQL QUERY

```

SELECT
  address_state AS State,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY address_state
ORDER BY address_state;

```

O U T P U T

	State	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	AK	78	1031800	1108570
	AL	432	4949225	5492272
	AR	236	2529700	2777875
	AZ	833	9206000	10041986
	CA	6894	78484125	83901234
	CO	770	8976000	9845810
	CT	730	8435575	9357612
	DC	214	2652350	2921854
	DE	110	1138100	1269136
	FL	2773	30046125	31601905
	GA	1355	15480325	16728040
	HI	170	1850525	2080184
	IA	5	56450	64482
	ID	6	59750	65329
	IL	1486	17124225	18875941
	IN	9	86225	85521
	KS	260	2872325	3247394
	KY	320	3504100	3792530
	LA	426	4498900	5001160
	MA	1310	15051000	16676279
	MD	1027	11911400	12985170
	ME	3	9200	10808
	MI	685	7829900	8543660
	MN	592	6302600	6750746
	MO	660	7151175	7692732
	MS	19	139125	149342
	MT	79	829525	892047
	NC	759	8787575	9534813
	NE	5	31700	24542
	NH	161	1917900	2101386
	NJ	1822	21657475	23425159
	NM	183	1916775	2084485
	NV	482	5307375	5451443
	NY	3701	42077050	46108181
	OH	1188	12991375	14330148
	OK	293	3365725	3712649
	OR	436	4720150	4966903
	PA	1482	15826525	17462908
	RI	196	1883025	2001774
	SC	464	5080475	5462458
	SD	63	606150	656514
	TN	17	162175	141522
	TX	2664	31236650	34392715
	UT	252	2849225	2952412
	VA	1375	15982650	17711443
	VT	54	504100	534973
	WA	805	8855525	9531739
	WI	446	5070450	5485161
	WV	167	1830525	1991936
	WY	79	890750	1046050

B3. TERM

1 Term-wise Summary

SQL QUERY

```
SELECT
  term AS Term,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY term
ORDER BY term;
```

OUTPUT

	Term	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	36 months	28237	273041225	294709458
	60 months	10339	162715850	178361475

B4. EMPLOYEE LENGTH

1 Employee Length-wise Summary

SQL QUERY

```
SELECT
  emp_length AS Employee_Length,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY emp_length
ORDER BY emp_length;
```

OUTPUT

	Employee_Length	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	< 1 year	4575	44210625	47545011
	1 year	3229	32883125	35498348
	10+ years	8870	116115950	125871616
	2 years	4382	44967975	49206961
	3 years	4088	43937850	47551832
	4 years	3428	37600375	40964850
	5 years	3273	36973625	40397571
	6 years	2228	25612650	27908658
	7 years	1772	20811725	22584136
	8 years	1476	17558950	19025777
	9 years	1255	15084225	16516173

B5. PURPOSE

1 Purpose-wise Summary

SQL QUERY

```
SELECT
  purpose AS PURPOSE,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY purpose
ORDER BY purpose;
```

OUTPUT

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	car	1497	10223575	11324914
	credit card	4998	58885175	65214084
	Debt consolidation	18214	232459675	253801871
	educational	315	2161650	2248380
	home improvement	2876	33350775	36380930
	house	366	4824925	5185538
	major purchase	2110	17251600	18676927
	medical	667	5533225	5851372
	moving	559	3748125	3999899
	other	3824	31155750	33289676
	renewable_energy	94	845750	898931
	small business	1776	24123100	23814817
	vacation	352	1967950	2116738
	wedding	928	9225800	10266856

B6. HOME OWNERSHIP

1 Home Ownership-wise Summary

SQL QUERY

```
SELECT
  home_ownership AS Home_Ownership,
  COUNT(id) AS Total_Loan_Applications,
  SUM(loan_amount) AS Total_Funded_Amount,
  SUM(total_payment) AS Total_Amount_Received
FROM financial_loan
GROUP BY home_ownership
ORDER BY home_ownership;
```

OUTPUT

	Home_Ownership	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	MORTGAGE	17198	219329150	238474438
	NONE	3	16800	19053
	OTHER	98	1044975	1025257
	OWN	2838	29597675	31729129
	RENT	18439	185768475	201823056

NOTE

Multiple filters have been applied on all the dashboards. You can check the results for these filters too by modifying the query and comparing the results.

For example: see the results when Grade A is selected in the dashboard filters.

+ Filter Example | Grade A**SQL QUERY**

```
SELECT
  purpose                AS PURPOSE,
  COUNT(id)              AS Total_Loan_Applications,
  SUM(loan_amount)       AS Total_Funded_Amount,
  SUM(total_payment)     AS Total_Amount_Received
FROM financial_loan
WHERE grade = 'A'
GROUP BY purpose
ORDER BY purpose;
```

OUTPUT

	PURPOSE	Total_Loan_Applications	Total_Funded_Amount	Total_Amount_Received
▶	car	577	3629475	3805538
	credit card	1353	12636075	13339495
	Debt consolidation	3753	37216300	38822971
	educational	79	484000	515639
	home improvement	933	8359175	8744006
	house	91	916575	957878
	major purchase	796	5344575	5604259
	medical	197	1471850	1526882
	moving	164	1069450	1099875
	other	1024	7043175	7397982
	renewable_energy	29	224150	225827
	small business	334	3172075	3190467
	vacation	122	683625	694542
	wedding	237	2001725	2126202